

# ✓ Most Important Data Engineering Concepts for Interviews

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## 1. ETL vs ELT

### You should know:

- Difference between ETL (transform before load) and ELT (transform in warehouse).
- Why modern systems prefer ELT (Snowflake/BigQuery power).
- When ETL is still required (Data cleansing before loading).

### Interview Q examples:

- “Explain ETL vs ELT and when to choose which.”
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## 2. Data Ingestion Patterns

### Batch ingestion

(e.g., Cloud Storage → BigQuery via scheduled loads)

### Streaming ingestion

(e.g., Pub/Sub → Dataflow → BigQuery)

### Near real-time ingestion

### Interview Q examples:

- “How do you design a system to ingest millions of events per second?”
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## 3. Change Data Capture (CDC)

Know multiple approaches:

- Log-based CDC (Debezium, Datastream)
- Timestamp-based CDC
- Hash-based CDC
- BigQuery MERGE for CDC

Interview Q examples:

- “How do you capture only updated rows from a source system?”
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## 4. Data Modeling Concepts

Be very familiar with:

- Star Schema (Facts + Dimensions)
- Snowflake schema
- Normalization vs Denormalization
- Slowly Changing Dimensions (SCD Type 1/2/3)

Interview Q examples:

- “What are SCD Types and how do you implement them?”

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## 5. Partitioning & Clustering

Particularly for **BigQuery**:

- Partitioning by ingestion time or date/timestamp
- Clustering on high cardinality columns
- How partitions improve cost and performance

**Interview Q examples:**

- “How does BigQuery reduce cost using partitioning?”
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## 6. Data Pipelines & Workflows

Understand:

- Orchestration tools
  - Airflow, Cloud Composer, Dagster, Prefect
- Pipeline components
  - ingestion → processing → quality → storage → serving

**Interview Q examples:**

- “How do you design an end-to-end data pipeline?”
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## 7. Distributed Processing Frameworks

### **Most important ones:**

- Apache Spark (must know)
- Apache Beam (GCP Dataflow)
- Kafka Streams / Flink (stream processing)

Know:

- RDD vs DataFrame
- Shuffle operations
- Spark partitions
- Lazy evaluation

### **Interview Q examples:**

- “Explain how Spark handles distributed joins.”
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## **8. Storage Systems**

### **OLTP vs OLAP**

### **Columnar vs row-based**

### **Data lake vs data warehouse vs data mart**

### **File formats:**

- Parquet (most important)
- Avro
- ORC

- JSON, CSV

#### Interview Q examples:

- “Why is Parquet preferred in data engineering pipelines?”
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## 9. SQL for Data Engineering

You *must* know:

- Window functions
- CTEs
- Aggregations
- Complex JOINS
- MERGE statements
- Error handling

#### Interview Q examples:

- “Write a query to get the top 3 products by sales per region.”
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## 10. Data Lake Architecture

#### Concepts to know:

- Bronze (raw)
- Silver (cleaned)

- Gold (business-ready)
- Delta Lake / Iceberg / Hudi

#### Interview Q examples:

- “Explain bronze-silver-gold architecture.”
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## 11. Data Quality & Validation

#### Tools/concepts:

- Great Expectations / Deequ
- Schema enforcement
- Constraints (null, type, ranges)

#### Interview Q examples:

- “How do you ensure data quality in a pipeline?”
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## 12. Workflow Reliability & Scalability Concepts

- Idempotency
- Backpressure
- Checkpointing
- Retry logic

- Reprocessing strategies
- Fault tolerance

**Interview Q examples:**

- “What happens if your pipeline fails in the middle of processing?”
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## 13. Cloud Concepts (GCP-focused)

Understand:

**For GCP:**

- BigQuery architecture (serverless, columnar, storage engine, slots)
- Dataflow (Apache Beam)
- Pub/Sub
- Cloud Composer (Airflow)
- Dataproc (Spark)
- Datastream (CDC)
- Cloud Storage (GCS)

**Interview Q examples:**

- “How does BigQuery achieve high performance without indexes?”
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## 14. Data Governance & Security

Know:

- Data lineage
- Data cataloging
- GDPR/PII handling
- Encryption at rest / in transit
- IAM roles

**Interview Q examples:**

- “How do you secure a data pipeline end-to-end?”
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## 15. Performance Optimization

Know techniques:

- SQL query optimization
- Using partition filters
- Avoiding cross joins
- Caching in Spark
- Choosing correct file formats

**Interview Q examples:**

- “How do you optimize BigQuery costs?”